BACKEND DEVELOPMENT WITH NODE.JS

- Project Title : Product inventory management
- Team Members :
- Aafila Parveen. I(Tl): Coding
- Bhuvaneshwari.H:Document Creation
- Asma Khatoon.g:Github Creation
- Mary jenniliya.p:Voice Note And Screen Recording

PROJECT INTRODUCTION:

Product Inventory Management is a system used to manage and track the products available in a store or organization. It helps businesses keep records of product details such as product name, quantity, price, and stock availability. Managing inventory manually can lead to errors, loss of data, and difficulty in tracking products.

The Product Inventory Management system is designed to make inventory management easier and more efficient. It allows users to add, update, delete, and view product details in a structured way. This system helps businesses maintain accurate stock information, avoid product shortages, and improve overall productivity.

By using this system, organizations can save time, reduce manual work, and manage their products in a more organized manner.

2. Objective

The main objectives of this project are:

1.To develop a system that manages product inventory efficiently.

2.To store and maintain product details such as name, quantity, and price.

3.To reduce manual errors in inventory management.

4.To make it easy to add, update, delete, and view product information.

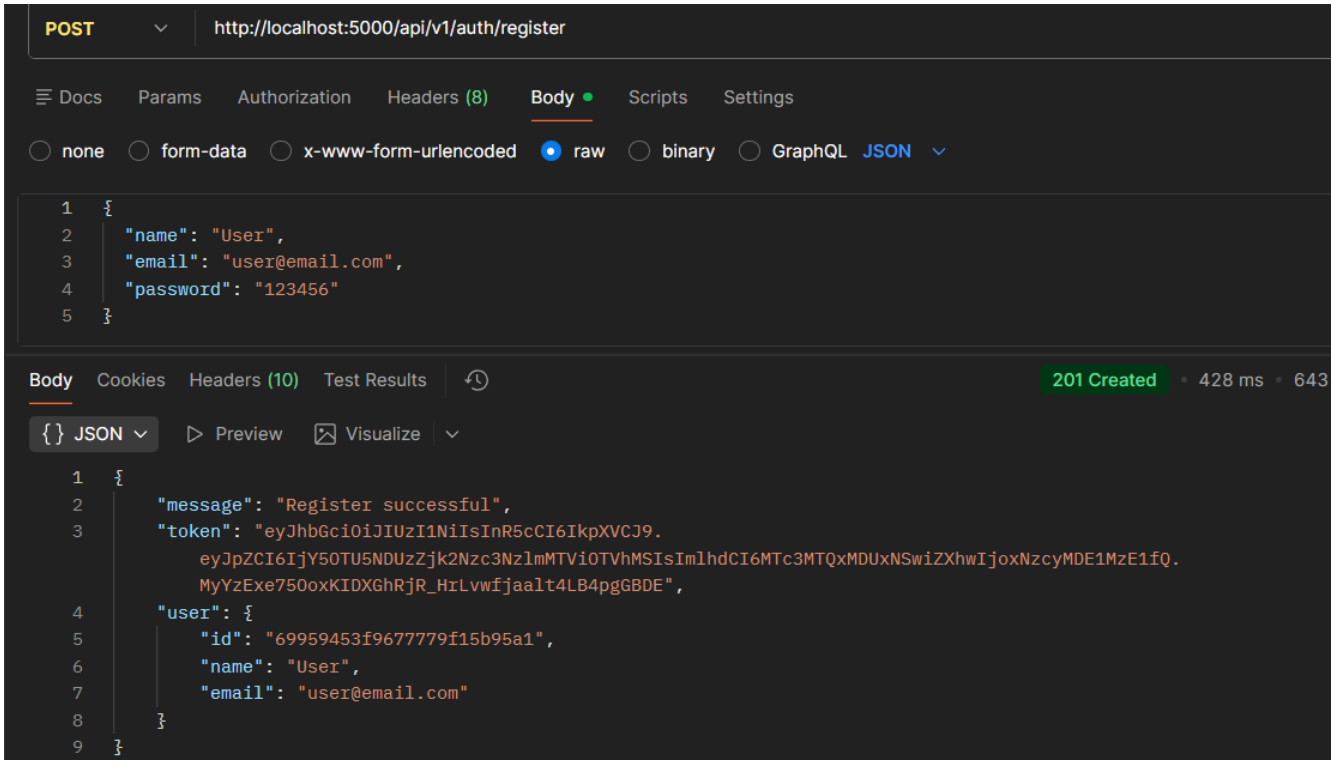
5.To help users track product availability and stock levels.

6.To improve the overall efficiency of inventory handling in an organization.

API IMPLEMENTATION SCREENSHOTS

The following screenshots demonstrate the successful execution of contact manager APIs using Thunder client in visual studio code, showing the request body and corresponding server responses.

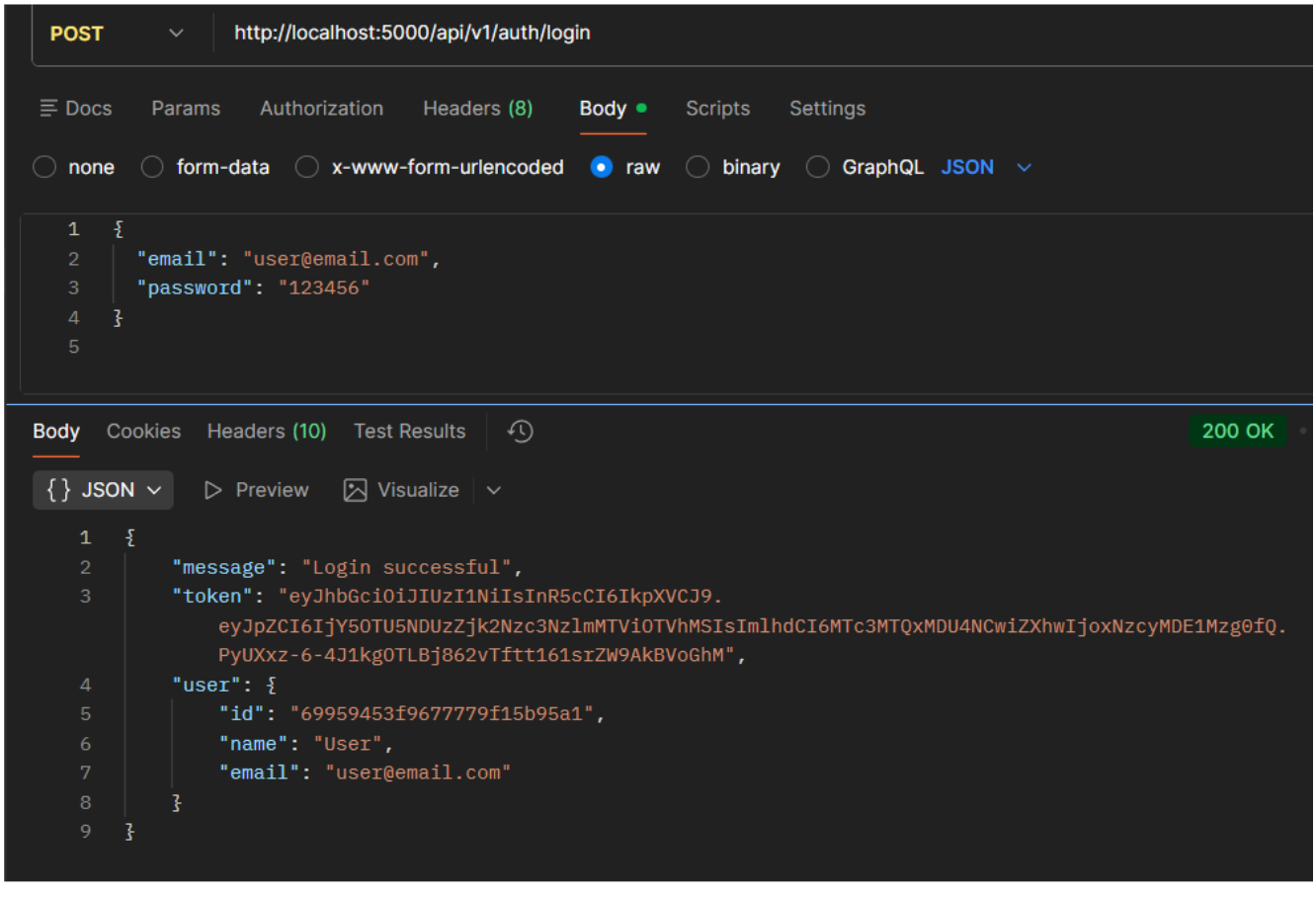
1. REGISTER API



- User Registration API: The screen shows a Register API being tested using a tool (like Postman/Thunder Client) with a POST request to.
- Request Data: The request body contains JSON user details including name, email, and password used to create a new account.

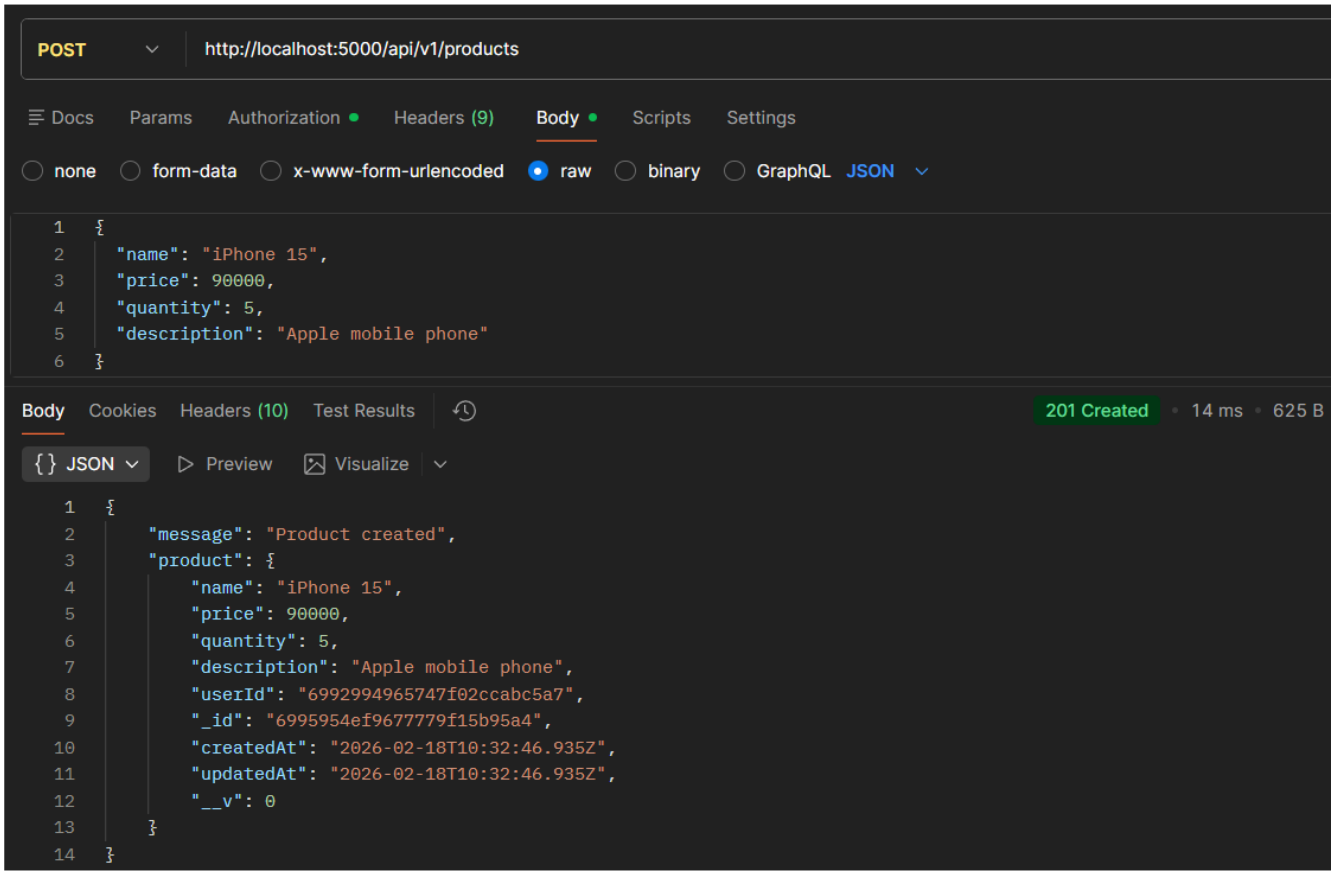
- Successful Response: The API returns a 201 Created status, along with a JWT token and user information (ID, name, email), indicating the registration was successful like Postman/Thunder Client) with a POST request to

- 2. LOGIN API



- Login API Request: The image shows a POST request to http://localhost:5000/api/v1/auth/login used to authenticate a user.
- User Credentials: The request body contains JSON data with email and password fields required for user login.
- Successful Response: The API returns 200 OK with a message “Login successful”, along with a JWT token and user details (ID, name, email

3.Create New Contact

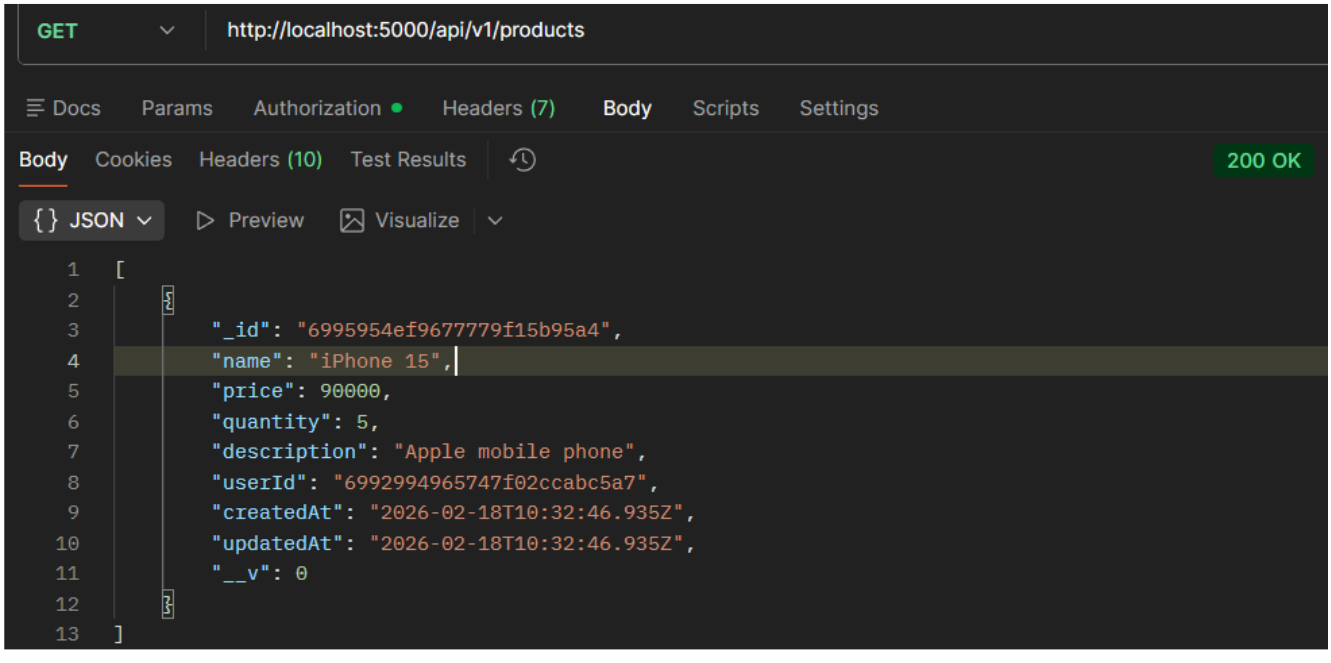


- Create Product API: The image shows a POST request to http://localhost:5000/api/v1/products used to add a new product to the system.
- Product Details: The request body contains JSON data including name, price, quantity, and description of the product.
- Successful Response: The API returns 201 Created with the message “Product created” and displays the saved product details such as product name, price, quantity, userId, and timestamps.

4.Create Product:

Allows an authenticated user to add a new product to their inventory by submitting product details. The product is saved with the user's ID to maintain ownership.

Get all Products



Get all Products:

Fetches all products that belong to the logged-in user. It ensures users can view only their own inventory data.

5.Update Product

PUT http://localhost:5000/api/v1/products/6995954ef9677779f15b95a4

Docs Params Authorization Headers (9) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
{
  "name": "iPhone 19 Pro",
  "price": 95000,
  "quantity": 10,
  "description": "Updated phone"
}
```

Body Cookies Headers (10) Test Results 200 OK 37 ms

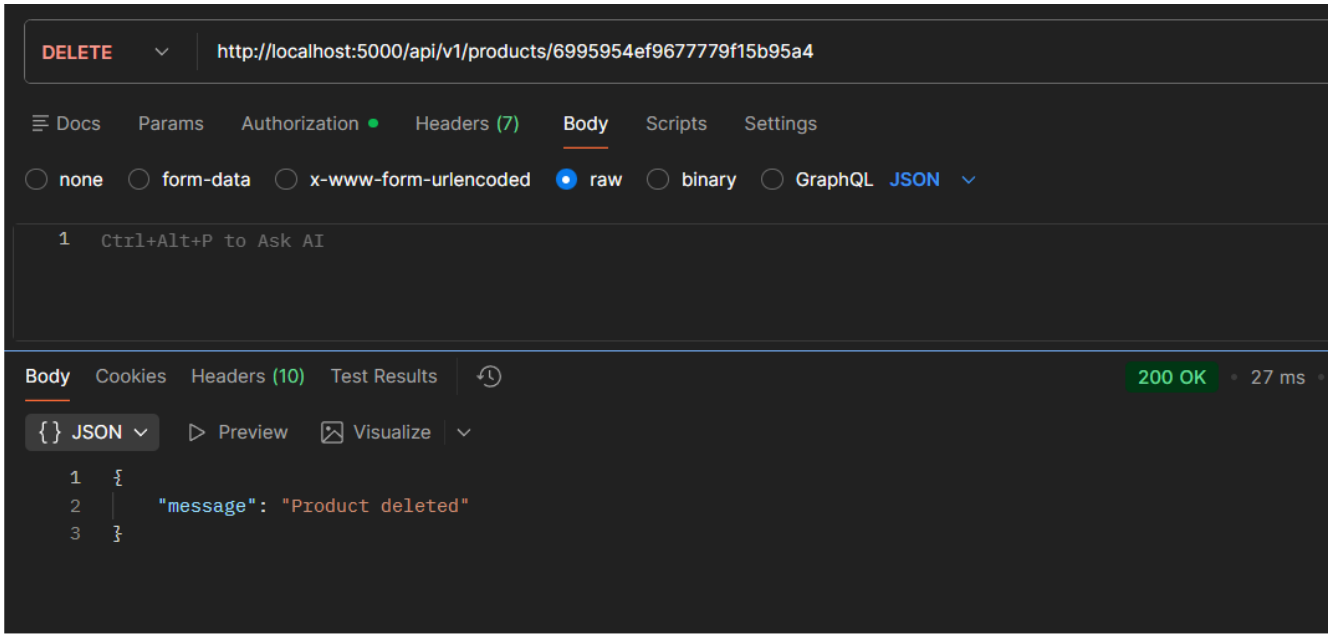
JSON Preview Visualize


```
{
    "message": "Product updated",
    "product": {
        "_id": "6995954ef9677779f15b95a4",
        "name": "iPhone 19 Pro",
        "price": 95000,
        "quantity": 10,
        "description": "Updated phone",
        "userId": "6992994965747f02ccabc5a7",
        "createdAt": "2026-02-18T10:32:46.935Z",
        "updatedAt": "2026-02-18T10:35:11.547Z",
        "__v": 0
    }
}
```


Update Product:

Enables users to modify existing product details such as name, price, or quantity. The system verifies ownership before allowing any updates.

6.Delete Product



Delete Product:

Allows users to permanently remove a product from their inventory. The operation is protected to ensure only the product owner can delete it.